

Manual Change History (3666410) QSK45 and QSK60 CENSE™

Electronic Control System Troubleshooting and Repair Manual

Date	Not Available
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Procedure Number	3666410-toc (/qs3/pubsys2/xml/en/manual/3666410/3666410-toc.html)
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Update Reason	

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Date	03-Apr-2003
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Update Reason	

Date	26-Aug-2003
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Date	26-Aug-2003
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Date	26-Aug-2003
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Update Reason	

Date	26-Aug-2003
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Update Reason	
Date	26-Aug-2003
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Update Reason	

Date	26-Aug-2003
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Section	Section TF - Troubleshooting Fault Codes
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Update Reason	
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Update Reason	
Date	26-Aug-2003
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Update Reason	

Date	26-Aug-2003
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Date	26-Aug-2003
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Update Reason	
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Update Reason	
Date	26-Aug-2003
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Date	26-Aug-2003
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Date	26-Aug-2003
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Date	26-Aug-2003
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Update Reason	
Date	26-Aug-2003
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Update Reason	
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Update Reason	

Date	26-Aug-2003
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Update Reason	
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Update Reason	
Date	27-Aug-2003
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Update Reason	
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Update Reason	
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Date	27-Jun-2006
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Update Reason	
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Update Reason	
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Update Reason	
Date	28-Jan-2008
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Update Reason	
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Update Reason	

Date	28-Feb-2008
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Update Reason	

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Update Reason	

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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	

Date	28-Feb-2008
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Update Reason	
Date	28-Feb-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc717 (/qs3/pubsys2/xml/en/procedures/81/81-fc717.html)
Procedure Title	High Power - Cylinder Number 7 Right Bank
Update Reason	
Date	28-Feb-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc718 (/qs3/pubsys2/xml/en/procedures/81/81-fc718.html)
Procedure Title	High Power - Cylinder Number 8 Right Bank
Update Reason	
Date	05-Mar-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc663 (/qs3/pubsys2/xml/en/procedures/81/81-fc663.html)
Procedure Title	High Power - Cylinder Number 3 Left Bank
Update Reason	
Date	05-Mar-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc713 (/qs3/pubsys2/xml/en/procedures/81/81-fc713.html)
Procedure Title	High Power - Cylinder Number 3 Right Bank
Update Reason	

Date	13-May-2008
Section	Section E - Engine and System Identification
Procedure Number	81-100-002 (/qs3/pubsys2/xml/en/procedures/81/81-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	

Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	
Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t030-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t030-tr.html)
Procedure Title	Cylinder Power is Low (Multiple Cylinders)
Update Reason	
Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t094-15-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t094-15-tr.html)
Procedure Title	Hot Shutdown
Update Reason	

Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t096-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t096-tr.html)
Procedure Title	Intake Manifold Air Temperature Above Specification
Update Reason	
Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t096-15-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t096-15-tr.html)
Procedure Title	Intake Manifold Air Temperature is High
Update Reason	
Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t096-20-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t096-20-tr.html)
Procedure Title	Intake Manifold Air Temperature is High (Rapid Rise)
Update Reason	
Date	04-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t103-55-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t103-55-tr.html)
Procedure Title	Lubricating Oil Filter Restriction is High
Update Reason	

Date	11-Jan-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t082-15-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t082-15-tr.html)
Procedure Title	Exhaust Temperature is High (Multiple Cylinders)
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	11-Mar-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t031-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t031-tr.html)
Procedure Title	Cylinder Power is Low (Single Cylinder)
Update Reason	

Date	11-Mar-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666410-t096-55-tr (/qs3/pubsys2/xml/en/manual/3666410/3666410-t096-55-tr.html)
Procedure Title	Intake Manifold Pressure is High
Update Reason	
Date	18-Mar-2010
Section	Section V - Specifications
Procedure Number	4021412 (/qs3/pubsys2/xml/en/procedures//4021412.html)
Procedure Title	Specifications (QSK60 Phase 2.0 Industrial with CENSE™ Wiring Diagram)
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	27-Apr-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc612 (/qs3/pubsys2/xml/en/procedures/81/81-fc612.html)
Procedure Title	High Lubricating Oil Filter Restriction
Update Reason	added to outline

Date	27-Apr-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-612 (/qs3/pubsys2/xml/en/procedures/81/81-t05-612.html)
Procedure Title	High Lubricating Oil Filter Restriction
Update Reason	added to outline
Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc621 (/qs3/pubsys2/xml/en/procedures/81/81-fc621.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 1 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.
Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-621 (/qs3/pubsys2/xml/en/procedures/81/81-t05-621.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 1 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.
Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc622 (/qs3/pubsys2/xml/en/procedures/81/81-fc622.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 3 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-622 (/qs3/pubsys2/xml/en/procedures/81/81-t05-622.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 3 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc623 (/qs3/pubsys2/xml/en/procedures/81/81-fc623.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 5 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-623 (/qs3/pubsys2/xml/en/procedures/81/81-t05-623.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 5 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc624 (/qs3/pubsys2/xml/en/procedures/81/81-fc624.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 7 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-624 (/qs3/pubsys2/xml/en/procedures/81/81-t05-624.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 7 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc625 (/qs3/pubsys2/xml/en/procedures/81/81-fc625.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 9 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-625 (/qs3/pubsys2/xml/en/procedures/81/81-t05-625.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 9 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc626 (/qs3/pubsys2/xml/en/procedures/81/81-fc626.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 11 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-626 (/qs3/pubsys2/xml/en/procedures/81/81-t05-626.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 11 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc627 (/qs3/pubsys2/xml/en/procedures/81/81-fc627.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 13 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-627 (/qs3/pubsys2/xml/en/procedures/81/81-t05-627.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 13 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc628 (/qs3/pubsys2/xml/en/procedures/81/81-fc628.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 15 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-628 (/qs3/pubsys2/xml/en/procedures/81/81-t05-628.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 15 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc631 (/qs3/pubsys2/xml/en/procedures/81/81-fc631.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 2 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-631 (/qs3/pubsys2/xml/en/procedures/81/81-t05-631.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 2 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc632 (/qs3/pubsys2/xml/en/procedures/81/81-fc632.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 4 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-632 (/qs3/pubsys2/xml/en/procedures/81/81-t05-632.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 4 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc633 (/qs3/pubsys2/xml/en/procedures/81/81-fc633.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 6 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-633 (/qs3/pubsys2/xml/en/procedures/81/81-t05-633.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 6 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc634 (/qs3/pubsys2/xml/en/procedures/81/81-fc634.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 8 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-634 (/qs3/pubsys2/xml/en/procedures/81/81-t05-634.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 8 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc635 (/qs3/pubsys2/xml/en/procedures/81/81-fc635.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 10 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-635 (/qs3/pubsys2/xml/en/procedures/81/81-t05-635.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 10 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc636 (/qs3/pubsys2/xml/en/procedures/81/81-fc636.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 12 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-636 (/qs3/pubsys2/xml/en/procedures/81/81-t05-636.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 12 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc637 (/qs3/pubsys2/xml/en/procedures/81/81-fc637.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 14 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-637 (/qs3/pubsys2/xml/en/procedures/81/81-t05-637.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 14 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc638 (/qs3/pubsys2/xml/en/procedures/81/81-fc638.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.

Date	02-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-638 (/qs3/pubsys2/xml/en/procedures/81/81-t05-638.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Added fault code to bulletin.
Date	31-May-2012
Section	Section 19 - Electronic Engine Controls
Procedure Number	81-022-001_19 (/qs3/pubsys2/xml/en/procedures/81/81-022-001_19.html)
Procedure Title	Service Tools
Update Reason	Added Service Tools: 3163064 (Bench Top Harness), 3885817 (In-Cab Data Link Cable), and 3825141 (In-Cab Data Link Cable).
Date	31-May-2012
Section	Section 19 - Electronic Engine Controls
Procedure Number	81-019-031 (/qs3/pubsys2/xml/en/procedures/81/81-019-031.html)
Procedure Title	Engine Control Module (ECM)
Update Reason	Revised Note in Install Step.
Date	3-Jul-2012
Section	Section F - Familiarization
Procedure Number	81-101-007 (/qs3/pubsys2/xml/en/procedures/81/81-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	Added Service Tools information: 3163064 (Bench Top Harness), 3885817 (In-Cab Data Link Cable), and 3825141 (In-Cab Data Link Cable).

Date	18-Jan-2013
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	81-019-431 (/qs3/pubsys2/xml/en/procedures/81/81-019-431.html)
Procedure Title	CENSE™ Datalogger
Update Reason	Added information concerning the CM2350 CENSE™ module.

Date	3-Jul-2012
Section	Section F - Familiarization
Procedure Number	81-101-007 (/qs3/pubsys2/xml/en/procedures/81/81-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	Added information concerning the CM2350 CENSE™ module.

Date	31-Oct-2013
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	81-019-013 (/qs3/pubsys2/xml/en/procedures/81/81-019-013.html)
Procedure Title	Exhaust Temperature Sensor
Update Reason	Brought the procedure up to date.

Date	17-Apr-2014
Section	Section 19 - Electronic Controls
Procedure Number	81-019-031 (/qs3/pubsys2/xml/en/procedures/81/81-019-031.html)
Procedure Title	Engine Control Module
Update Reason	Revise jackscrew torque value and added a note to not over torque.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc621 (/qs3/pubsys2/xml/en/procedures/81/81-fc621.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 1 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc622 (/qs3/pubsys2/xml/en/procedures/81/81-fc622.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 3 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to the location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc623 (/qs3/pubsys2/xml/en/procedures/81/81-fc623.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 5 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc624 (/qs3/pubsys2/xml/en/procedures/81/81-fc624.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 7 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc625 (/qs3/pubsys2/xml/en/procedures/81/81-fc625.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 9 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc626 (/qs3/pubsys2/xml/en/procedures/81/81-fc626.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 11 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc627 (/qs3/pubsys2/xml/en/procedures/81/81-fc627.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 13 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc628 (/qs3/pubsys2/xml/en/procedures/81/81-fc628.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 15 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc631 (/qs3/pubsys2/xml/en/procedures/81/81-fc631.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 2 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc632 (/qs3/pubsys2/xml/en/procedures/81/81-fc632.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 4 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc633 (/qs3/pubsys2/xml/en/procedures/81/81-fc633.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 6 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc634 (/qs3/pubsys2/xml/en/procedures/81/81-fc634.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 8 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc635 (/qs3/pubsys2/xml/en/procedures/81/81-fc635.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 10 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc636 (/qs3/pubsys2/xml/en/procedures/81/81-fc636.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 12 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc637 (/qs3/pubsys2/xml/en/procedures/81/81-fc637.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 14 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc638 (/qs3/pubsys2/xml/en/procedures/81/81-fc638.html)
Procedure Title	Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level
Update Reason	Updated circuit art and reference to location of Engine Performance Troubleshooting Tree in the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc661 (/qs3/pubsys2/xml/en/procedures/81/81-fc661.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 1 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc662 (/qs3/pubsys2/xml/en/procedures/81/81-fc662.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 3 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc663 (/qs3/pubsys2/xml/en/procedures/81/81-fc663.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 5 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc664 (/qs3/pubsys2/xml/en/procedures/81/81-fc664.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 7 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc665 (/qs3/pubsys2/xml/en/procedures/81/81-fc665.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 9 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc666 (/qs3/pubsys2/xml/en/procedures/81/81-fc666.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 11 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc667 (/qs3/pubsys2/xml/en/procedures/81/81-fc667.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 13 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc668 (/qs3/pubsys2/xml/en/procedures/81/81-fc668.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 15 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc671 (/qs3/pubsys2/xml/en/procedures/81/81-fc671.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 1 (A1) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc672 (/qs3/pubsys2/xml/en/procedures/81/81-fc672.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 3 (A2) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc673 (/qs3/pubsys2/xml/en/procedures/81/81-fc673.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 5 (A3) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc674 (/qs3/pubsys2/xml/en/procedures/81/81-fc674.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 7 (A4) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc675 (/qs3/pubsys2/xml/en/procedures/81/81-fc675.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 9 (A5) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc676 (/qs3/pubsys2/xml/en/procedures/81/81-fc676.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 11 (A6) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc677 (/qs3/pubsys2/xml/en/procedures/81/81-fc677.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 13 (A7) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc678 (/qs3/pubsys2/xml/en/procedures/81/81-fc678.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 15 (A8) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc711 (/qs3/pubsys2/xml/en/procedures/81/81-fc711.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 2 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc712 (/qs3/pubsys2/xml/en/procedures/81/81-fc712.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 4 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc713 (/qs3/pubsys2/xml/en/procedures/81/81-fc713.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 6 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc714 (/qs3/pubsys2/xml/en/procedures/81/81-fc714.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 8 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc715 (/qs3/pubsys2/xml/en/procedures/81/81-fc715.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 10 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc716 (/qs3/pubsys2/xml/en/procedures/81/81-fc716.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 12 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc717 (/qs3/pubsys2/xml/en/procedures/81/81-fc717.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 14 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc718 (/qs3/pubsys2/xml/en/procedures/81/81-fc718.html)
Procedure Title	Exhaust Gas Temperature Deviation High for Cylinder 16 - Data Valid But Above Normal Operating Range - Moderately Severe Level
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc721 (/qs3/pubsys2/xml/en/procedures/81/81-fc721.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 2 (B1) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc722 (/qs3/pubsys2/xml/en/procedures/81/81-fc722.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 4 (B2) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc723 (/qs3/pubsys2/xml/en/procedures/81/81-fc723.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 6 (B3) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc724 (/qs3/pubsys2/xml/en/procedures/81/81-fc724.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 8 (B4) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc725 (/qs3/pubsys2/xml/en/procedures/81/81-fc725.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 10 (B5) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc726 (/qs3/pubsys2/xml/en/procedures/81/81-fc726.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 12 (B6) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc727 (/qs3/pubsys2/xml/en/procedures/81/81-fc727.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 14 (B7) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.

Date	10-Jul-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc728 (/qs3/pubsys2/xml/en/procedures/81/81-fc728.html)
Procedure Title	Exhaust Gas Temperature Sensor Circuit Cylinder 16 (B8) - Voltage Below Normal or Shorted to Low Source
Update Reason	Updated circuit art and the Shoptalk.
Date	12-Aug-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc1117 (/qs3/pubsys2/xml/en/procedures/81/81-fc1117.html)
Procedure Title	Power Supply Lost With Ignition ON - Data Erratic, Intermittent, or Incorrect
Update Reason	Added fault code to manual.
Date	13-Aug-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-t05-1117 (/qs3/pubsys2/xml/en/procedures/81/81-t05-1117.html)
Procedure Title	Power Supply Lost With Ignition ON - Data Erratic, Intermittent, or Incorrect
Update Reason	Added fault code to manual.
Date	28-Oct-2022
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	81-fc612 (/qs3/pubsys2/xml/en/procedures/81/81-fc612.html)
Procedure Title	Engine Oil Filter Differential Pressure - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Updated listed effect of fault code to note that possible reduced engine performance can occur.

